

NASIK, India

WMO: 429210

Lat: 19.967N

Lon: 73.817E

Elev: 598

StdP: 94.34

Time Zone: 5.50 (IND)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.2	10.6	-1.1	3.7	28.2	0.8	4.3	27.5	3.4	26.2	3.1	25.4	0.0	320	0.254

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.2	38.2	19.5	37.3	19.5	36.2	19.6	25.0	29.7	24.6	29.0	24.2	28.3	1.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	20.2	26.5	23.6	19.8	26.0	23.2	19.3	25.6	80.1	29.8	78.4	29.1	76.8	28.5	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.4	3.5	3.2	6.5	40.2	1.5	0.9	5.5	40.9	4.6	41.4	3.7	41.9	2.7	42.5		
			4.7	26.5	2.0	0.9	3.3	27.2	2.1	27.7	1.0	28.3	-0.4	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.7	19.8	22.3	25.9	29.0	29.7	27.6	24.9	24.3	24.9	24.9	22.7	20.7
	DBStd	3.40	1.96	2.30	2.39	1.74	1.13	1.75	1.17	0.94	1.13	1.47	1.85	2.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	17	9	3	0	0	0	0	0	0	0	0	0	5
	CDD10.0	5374	304	344	492	570	611	528	461	444	448	462	381	331
	CDD18.3	2351	54	114	233	320	352	278	202	186	198	203	131	78
	CDH23.3	22875	746	1336	2906	4273	4474	2648	962	691	1077	1729	1222	812
	CDH26.7	10337	181	589	1657	2622	2572	1012	131	61	238	683	399	194
Wind	WSAvg	1.0	0.5	0.7	0.8	1.1	1.7	1.7	1.5	1.3	0.9	0.6	0.6	0.6
Precipitation	PrecAvg	682	2	0	2	4	17	102	185	139	164	44	21	8
	PrecMax	1145	24	11	19	44	98	293	437	338	426	140	108	97
	PrecMin	388	0	0	0	0	0	0	64	44	17	0	0	0
	PrecStd	171	6	2	5	10	27	83	97	76	107	43	31	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	34.7	38.3	39.7	39.2	36.2	30.9	29.3	31.7	33.1	31.7	31.1
		MCWB	17.0	17.2	18.1	18.8	21.1	23.6	23.7	24.1	23.7	20.8	19.3	18.4
	2%	DB	29.6	33.2	36.9	38.5	38.1	34.7	29.3	28.2	30.3	32.1	30.9	29.8
		MCWB	16.7	16.9	18.0	18.9	20.6	23.3	23.6	23.7	23.7	20.9	19.3	18.2
	5%	DB	28.7	31.8	35.5	37.6	37.1	33.2	28.1	27.3	29.1	31.2	30.1	28.7
		MCWB	16.4	16.7	17.5	18.9	20.7	23.5	23.5	23.3	23.6	20.5	19.1	17.9
	10%	DB	27.6	30.5	34.3	36.5	36.0	31.7	27.1	26.5	28.0	30.4	29.1	27.7
		MCWB	16.2	16.5	17.5	18.8	21.1	23.7	23.4	23.1	23.3	20.5	19.1	17.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	20.0	22.6	23.2	24.9	25.6	24.9	24.8	25.7	25.0	23.5	21.7
		MCDB	26.9	29.8	31.9	32.1	33.5	30.9	28.1	27.8	29.0	28.9	28.2	27.4
	2%	WB	18.4	18.9	20.4	21.9	24.0	25.0	24.2	24.2	24.7	24.2	22.8	20.5
		MCDB	26.4	28.8	30.6	30.4	31.9	30.4	27.1	26.9	28.2	28.1	27.3	26.6
	5%	WB	17.7	18.0	19.3	21.1	23.6	24.6	23.9	23.7	24.2	23.6	22.1	19.7
		MCDB	26.1	27.5	30.3	30.4	30.9	29.7	26.6	26.2	27.4	27.4	26.0	25.9
	10%	WB	16.9	17.3	18.5	20.5	23.1	24.2	23.6	23.4	23.7	23.1	21.4	18.8
		MCDB	25.4	27.4	29.9	30.5	30.0	29.0	26.1	25.6	26.6	26.7	25.2	25.1
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.4	16.5	15.6	13.2	7.6	4.2	4.2	6.4	11.3	13.8	15.0
		MCDBR	17.0	17.8	17.5	16.4	14.8	10.4	6.1	5.7	8.3	13.1	14.7	15.9
		MCWBR	6.1	5.3	4.7	4.3	4.1	2.2	2.1	2.3	3.2	4.0	5.1	6.0
	5% WB	MCDBR	14.5	15.2	15.3	15.0	12.3	8.1	4.6	4.9	6.9	8.9	9.4	12.8
	MCWBR	5.9	5.2	5.3	4.8	3.5	2.7	2.1	2.3	3.2	3.6	4.0	5.5	
Clear-Sky Solar Irradiance	taub	0.418	0.423	0.470	0.510	0.526	0.595	0.585	0.548	0.530	0.531	0.485	0.444	
	taud	2.159	2.152	2.032	1.971	1.972	1.862	1.895	1.998	2.029	1.954	2.027	2.088	
	Ebn at Noon	849	870	842	810	789	729	738	770	779	759	774	805	
	Edh at Noon	140	149	175	188	186	205	199	180	173	180	159	147	
All-Sky Solar Radiation	RadAvg	4.83	5.68	6.58	7.13	7.15	4.96	3.44	3.48	4.43	5.21	4.83	4.60	
	RadStd	0.14	0.17	0.21	0.15	0.31	0.50	0.43	0.28	0.40	0.28	0.26	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.36	+1.01	N/A	N/A	N/A	+0.21	N/A	N/A	+160

Nomenclature: See separate page